Notes: Azoulay, Graff Zivin, and Wang (QJE, 2010)

Superstar Extinction

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1 Big picture

- **Question.** Do scientific superstars generate spillovers for their collaborators? The paper asks whether collaboration with an eminent scientist raises coauthors' productivity, and whether the effect works through ideas rather than merely through resources or social access.
- **Setting.** The paper studies academic life sciences, where collaboration leaves a detailed paper trail through coauthored publications, citations, PubMED records, NIH grants, and medical-school faculty data.
- **Natural experiment.** The central source of variation is the premature and unexpected death of 112 active superstar life scientists. A sudden death removes a collaborator, mentor-like peer, source of ideas, and network node in a way that is plausibly not chosen by the surviving coauthors.
- **Main outcome.** The core outcome is collaborators' quality-adjusted publication output, measured by journal-impact-factor-weighted publications. The authors also distinguish publications written with the superstar from publications written with other coauthors.
- **Main result.** After the death of a superstar, collaborators experience a lasting decline of roughly 5% to 8% in quality-adjusted publication rates.
- **Mechanism question.** The paper asks whether this drop reflects imperfect replacement of a missing team member, loss of access to resources and gatekeeping, or genuine knowledge spillovers from the superstar.
- **Mechanism result.** The evidence favors knowledge spillovers. Coauthors who are intellectually close to the star suffer larger losses, while simple measures of geographic proximity, NIH gatekeeping, network centrality, and former trainee status do not explain the main result.
- **Economic takeaway.** Scientific collaboration creates idea-based externalities that are difficult to replace. When a superstar dies, part of the local scientific field - the "invisible college" around that star - contracts as well.

2 Incremental contribution of the paper

- **From collaboration correlations to causal spillovers.** Earlier work documents that science is increasingly team-based, but coauthorship is endogenously formed. This paper uses sudden superstar deaths to study the causal effect of losing a collaborator.
- **A novel "superstar extinction" research design.** The paper treats premature and unexpected deaths as shocks to coauthorship networks. This design provides variation in exposure to superstar talent that is less contaminated by matching and selection.
- **A large-scale individual-level data construction.** The authors combine an elite scientist sample, AAMC Faculty Roster data, PubMED publication histories, NIH funding data, citation data, and software-based collaborator matching.
- **A matched control contribution.** The paper constructs matched control collaborators from coauthors of superstars who do not die, using coarsened exact matching on career age, prior output, collaboration intensity, collaboration recency, and star citation status.

- **A mechanism contribution.** The paper does not stop at showing a productivity decline. It uses dyad characteristics - coauthorship intensity, recency, MeSH keyword overlap, colocation, "accidental" coauthorship, former trainee links, NIH review-panel ties, and star status - to distinguish ideas from resources.
- **A concept of spillovers in idea space.** The key conclusion is that spillovers are more bounded by intellectual proximity than by physical location or formal access to resources. This gives empirical content to the older notion of an invisible college.
- **A policy-relevant contribution.** If superstars generate spillovers that are not fully internalized, then scientific labor markets, grant allocation, and team formation may underprovide the social value of collaborations with especially creative researchers.

3 Data and measurement

3.1 Superstar sample

- The authors begin with a broad elite sample of 10,349 academic life scientists, roughly the top 5% of the relevant labor market.
- A scientist enters the elite sample if he or she satisfies at least one of seven criteria: highly funded by NIH, highly cited, top patenting, National Academy of Sciences membership, NIH MERIT award, Howard Hughes investigator status, or early-career prize winner.
- The treatment sample is restricted to elite scientists who die between 1979 and 2003, die at age 67 or below, remain research-active near death, and die suddenly and unexpectedly.
- These restrictions produce 112 extinct superstars.
- The average superstar dies at about age 57, has 47 collaborators, publishes about 140 career papers, accumulates more than 8,000 citations, and receives more than $10 million in career NIH funding.

Interpretation: The superstar sample is designed to capture scientists whose ideas and networks could plausibly matter for others. The sudden-death restriction is essential because expected decline, retirement, or illness could otherwise contaminate the estimates.

3.2 Collaborator and control samples

- The paper uses the Association of American Medical Colleges Faculty Roster from 1975 to 2006 to identify faculty members in U.S. medical schools.
- Publications are obtained from PubMED. The authors develop the Stars/Colleague Generator, S/CGEN, to download publications, strip coauthors, match coauthors to the faculty roster, and construct yearly publication histories.
- The software identifies 5,267 distinct treated collaborators with unique PubMED names for the 112 deceased superstars.

- For 5,064 treated collaborators, the paper identifies a matched control collaborator from the coauthors of elite scientists who do not die prematurely.
- The final matched sample contains 10,128 treated and control collaborators.

Interpretation: The collaborator-level panel is the empirical backbone of the paper. It turns a qualitative shock - the loss of a superstar - into a repeated yearly productivity panel for treated and matched scientists.

3.3 Output measures

- The main outcome is the number of publications produced by collaborator j in year t.
- Publications are quality-adjusted using journal impact factors, so the main dependent variable is JIF-weighted publications.
- The authors also compute output excluding publications coauthored with the deceased superstar.
- The distinction is important because it separates the mechanical loss of joint work with the star from broader productivity declines in the collaborator's independent or alternative collaborative output.

Interpretation: If only star-coauthored papers disappear, the result would be a mechanical team-composition effect. A decline in publications written with others suggests a deeper loss of ideas, direction, or scientific momentum.

3.4 Dyad-level variables

- **Coauthorship intensity:** casual dyads have one or two coauthored papers, regular dyads have three to nine, and close dyads have ten or more.
- **Collaboration recency:** recent dyads have a joint publication in the three years before the star's death.
- **Former trainee status:** a coauthor is classified as a former trainee when authorship position and degree timing indicate graduate or postdoctoral training under the star.
- **Colocation:** the paper records whether the collaborator and star are at the same institution at the time of death.
- **Intellectual proximity:** the main measure is normalized overlap in MeSH keywords from noncoauthored publications in the five years before death.
- **Accidental coauthorship:** dyads are classified as especially weak social ties if the star and collaborator only appear together in the middle of the authorship list.

Interpretation: The dyad variables allow the paper to ask what kind of proximity matters: shared projects, recent work, physical location, social contact, or similarity of scientific ideas.

3.5 Star-level variables

- The paper measures whether the superstar sat on NIH review panels and how many of the star's coauthors sat on NIH panels.
- It also measures the star's betweenness centrality in the elite-scientist collaboration network.
- Superstar status is measured using cumulative citations and cumulative NIH funding.
- These variables are used to separate resource access and gatekeeping stories from idea-spillover stories.

Interpretation: If the superstar mainly matters as a gatekeeper, losses should be larger when the star controls funding access, sits in central network positions, or channels resources to former trainees. The paper finds little support for that view.

4 Models and empirical specifications

4.1 Baseline fixed-effects Poisson model

The main estimating equation relates collaborator j's output in year t to the death status of superstar i:

$$E[y_{jt} \mid X_{ijt}] = \exp\left[\beta_0 + \beta_1 AfterDeath_{it} + f(Age_{jt}) + \delta_t + \gamma_{ij}\right].$$

- y_{jt} is publication output, usually JIF-weighted publications.
- $AfterDeath_{it}$ equals one in years after superstar i dies.
- $f(Age_{jt})$ is a flexible set of career-age controls.
- δ_t are calendar-year effects.
- γ_{ij} are dyad fixed effects, so identification comes from within-dyad changes over time.

Interpretation: The coefficient β_1 measures how a collaborator's publication rate changes after the associated superstar dies, relative to the same collaborator-star dyad before death and relative to the age and calendar-time profile.

4.2 Poisson quasi-maximum likelihood

- Publication counts are nonnegative and skewed, with many low-output years.
- The paper therefore uses fixed-effect Poisson quasi-maximum likelihood.
- The Poisson QML approach remains consistent if the conditional mean is correctly specified, even if the outcome is not literally Poisson.
- Standard errors are robust and clustered at the superstar level.
- Coefficients can be interpreted approximately as log changes; for example, $1 - \exp(-0.092) = 8.79\%$.

Interpretation: The estimator is designed for count-like publication outcomes while preserving the panel structure and fixed effects.

4.3 Matched difference-in-differences framework

- The paper's preferred design adds matched control collaborators from the coauthors of elite scientists who do not die.
- Controls are selected using coarsened exact matching.
- Matching variables include collaborator degree year, number of coauthorships with the star, years since last coauthorship, publication flow in the year of death, cumulative publication stock before death, and the star's cumulative citation count.
- The matched controls help pin down age and calendar-time effects, and they reduce concern that treated collaborators are simply moving along a special collaboration life cycle.

Interpretation: The matched design asks whether treated collaborators decline relative to similar collaborators who had comparable career age, prior output, collaboration intensity, collaboration recency, and superstar status, but whose associated stars did not die.

4.4 Heterogeneity and mechanism regressions

The paper then interacts $AfterDeath_{it}$ with star, collaborator, and dyad characteristics:

$$E[y_{jt} \mid X_{ijt}] = \exp\left[\beta_0 + \beta_1 AfterDeath_{it} + \beta_2 AfterDeath_{it} \times Z_{ij} + f(Age_{jt}) + \delta_t + \gamma_{ij}\right].$$

- Z_{ij} can be coauthorship intensity, collaboration recency, colocation, keyword overlap, accidental coauthorship, former trainee status, network centrality, NIH-panel ties, star citations, or star funding.
- A negative interaction means that the post-death decline is larger for that dyad or star type.
- The mechanism tests ask whether the main effect is stronger in places predicted by skill-substitution, resource-access, or knowledge-spillover mechanisms.

Interpretation: The core empirical strategy estimates the average effect of losing a superstar. The interaction models explain why the effect occurs.

5 Identification and empirical design

5.1 What the design can identify

- The design identifies the effect of losing an active superstar collaborator on surviving coauthors' publication output.
- It is not an experiment in assigning collaborators to stars; rather, it is a quasi-experiment in removing stars from existing collaboration networks.

- The key identifying comparison is before versus after the sudden death, relative to matched controls and net of age and calendar-year effects.
- The result should be interpreted as the impact of losing the star, not the full social value of exposure to the star while alive.

Interpretation: The design is strong for estimating the consequence of superstar extinction. It cannot directly measure the full counterfactual scientific output that collaborators would have produced if the star had never existed.

5.2 Sudden death and no anticipation

- The paper restricts the treatment sample to deaths described as sudden and unexpected.
- It excludes deaths that could plausibly have been anticipated by collaborators.
- This matters because illness, retirement, or decline before death could already affect collaborator output and would blur the timing of the treatment.

Interpretation: The suddenness requirement makes the death closer to an exogenous shock to the coauthorship network.

5.3 Life-cycle and calendar-time confounds

- Scientific output varies with career age and calendar time.
- Collaborations may also have their own life cycle: projects begin, mature, and eventually fade.
- The baseline model controls flexibly for career age and year effects.
- The matched control group is designed to address the possibility that treated collaborators would have declined even without the superstar death.

Interpretation: The central threat is that a post-death decline could reflect aging, field decline, or a collaboration cycle rather than the missing superstar. The paper's matching and pretrend evidence are meant to address this.

5.4 Control group and spillover leakage

- Control collaborators are chosen from coauthors of elite scientists who do not die prematurely.
- A concern is that control scientists might also be indirectly affected by the death of a superstar through the broader scientific network.
- The paper checks robustness to possible leakage through the coauthorship network.
- Results are similar with and without the explicit control group, which reduces concern that the main result is an artifact of matching.

Interpretation: In a networked scientific labor market, no scientist is completely isolated. The paper therefore treats the control group as useful but not perfect, and it verifies that the main result does not hinge on it.

5.5 Mechanism identification

- The mechanism tests are not fully randomized experiments.
- They rely on whether the pattern of heterogeneous treatment effects matches the predictions of different mechanisms.
- Skill-substitution predicts larger effects for close, recent, and active collaborators.
- Gatekeeping predicts larger effects when stars control NIH resources, occupy central network positions, or maintain former trainees.
- Knowledge spillovers predict larger effects for collaborators close in intellectual space and for collaborators of especially scientifically influential stars.

Interpretation: The mechanism evidence is circumstantial but powerful because many resource-access predictions fail while idea-space predictions succeed.

6 Tables and figures

6.1 Table I: Summary Statistics for Superstar Scientists

Read:

- Table I describes the 112 superstars who die suddenly while still active in research.
- The average star dies at age 57 and received the most recent degree around 1963.
- Stars are extremely productive: the average has about 140 publications, 8,190 citations, and more than $10 million in NIH funding.
- The average superstar has 47 distinct collaborators.
- Only a small fraction sat on NIH review panels in the five years before death, which later helps evaluate gatekeeping stories.

Economic message: The treatment group is genuinely elite. The table establishes that the paper is not studying ordinary coauthors, but scientists with unusually high scientific achievement and broad collaborative reach.

TABLE I

SUMMARY STATISTICS FOR SUPERSTAR SCIENTISTS ($N = 112$)

	Mean	Median	Std. Dev.	Min.	Max.
Birth age at death	57.170	58	7.042	37	67
Degree year	1962.741	1964	10.193	1942	1984
M.D.	0.420	0	0.496	0	1
Ph.D.	0.438	0	0.498	0	1
M.D./Ph.D.	0.143	0	0.351	0	1
Female	0.063	0	0.243	0	1
U.S. born	0.786	1	0.412	0	1
No. of collaborators	47.027	37	34.716	3	178
NIH review panel membership (past 5 yrs)	0.045	0	0.207	0	1
No. of collabs. in NIH review panels (past 5 yrs)	1.330	1	1.657	0	7
Career no. of publications	139.607	121	91.371	25	473
Career no. of citations	8,190	6,408	7,593	435	38,941
Career NIH funding	$10,722,590	$8,139,397	$12,057,638	$0	$70,231,584

Notes. Sample consists of 112 superstar life scientists who died suddenly while still actively engaged in research. See Appendix I and Section II.A for details on sample construction. Degree year denotes the year of the most recent degree attained by the superstar. Number of collaborators is defined as the number of distinct coauthors within the scientists' cumulative stock of publications. NIH review panel membership denotes stars who were members of an NIH review panel in the five years prior to their death, and the number of collaborators in NIH review panels refers to the number of coauthors of each superstar who sat on NIH review panels in the five years prior to the star's death. We use the terms "star" and "superstar" interchangeably.

Table I: Summary Statistics for Superstar Scientists.

6.2 Table II: Summary Statistics for Collaborators

Read:

- Table II compares matched control collaborators and treated collaborators in the year of superstar death.
- The control and treated groups are closely balanced on publication flow, cumulative publication stock, career age, coauthorship intensity, years since first and last coauthorship, MeSH keyword overlap, and star citation count.
- Treated collaborators have an average of 19.1 JIF-weighted publications in the year of death; controls have 18.3.
- The average collaboration is not extremely close: median cumulative coauthorships equal one for both groups.
- About 8% of treated collaborators are former trainees of the star, and about 12% are colocated with the star.

Economic message: Table II supports the matched comparison. Treated and control collaborators look very similar on the core variables that matter for publication dynamics and collaboration history.

TABLE II

SUMMARY STATISTICS FOR COLLABORATORS IN THE YEAR OF SUPERSTAR DEATH

	Mean	Median	Std. dev.	Min.	Max.
	Control collaborators ($N = 5{,}064$)				
No. of weighted publications	18.314	8	27.917	0	342
Cum. no. of weighted publications	327.330	187	409.098	0	3,968
Holds R01 grant	0.559	1	0.497	0	1
Colocated	0.144	0	0.351	0	1
Career age	23.698	23	9.963	1	59
Elite	0.093	0	0.290	0	1
Cum. no. of coauthorships	2.734	1	4.339	1	69
No. of other superstar collaborators	2.746	2	3.516	0	31
Years since first coauthorship	10.949	10	7.901	0	42
Years since last coauthorship	9.275	8	7.774	0	41
Former trainee of the star	0.070	0	0.255	0	1
"Accidental" collaborator	0.076	0	0.265	0	1
MeSH keyword overlap	0.265	0	0.162	0	1
Superstar citation count	10,083	7,245	8,878	99	90,136

Table II, part 1: Control collaborators.

TABLE II

(CONTINUED)

	Mean	Median	Std. dev.	Min.	Max.
	Treated collaborators ($N = 5{,}064$)				
No. of weighted publications	19.068	8	31.656	0	491
Cum. no. of weighted publications	334.905	187	436.927	0	4,519
Holds R01 grant	0.571	1	0.495	0	1
Colocated	0.123	0	0.328	0	1
Career age	23.761	23	9.969	0	59
Elite	0.077	0	0.266	0	1
Cum. no. of coauthorships	2.835	1	4.894	1	75
No. of other superstar collaborators	3.087	2	4.255	0	44
Years since first coauthorship	11.022	10	7.896	0	39
Years since last coauthorship	9.255	8	7.728	0	38
Former trainee of the star	0.084	0	0.278	0	1
"Accidental" collaborator	0.075	0	0.264	0	1
MeSH keyword overlap	0.259	0	0.157	0	1
Superstar citation count	10,228	7,239	7,952	397	34,746

Notes. The samples consist of faculty collaborators of 112 deceased superstar life scientists and an equal number of matched control coauthors. See Sections II.B and III for details on the sample construction and variable definitions and Appendix II for details on the matching procedure. All variables are measured as of the year of superstar death. Publications are JIF-weighted.

Table II, part 2: Treated collaborators.

6.3 Figures I and II: Collaboration Intensity and Ideas-Space Proximity

Read:

- Figure I shows the distribution of coauthorship intensity. Most collaborators have only one or two coauthored papers with the superstar.
- This skewness motivates the paper's categories of casual, regular, and close collaborators.
- Figure II shows the distribution of normalized MeSH keyword overlap between the star and collaborator, excluding coauthored work.
- The keyword-overlap measure captures proximity in intellectual space rather than physical or social space.

Economic message: These figures show that collaboration is heterogeneous. Some coauthors have only a weak tie to the superstar, while others share a closely related intellectual agenda. That heterogeneity is central to the mechanism tests.

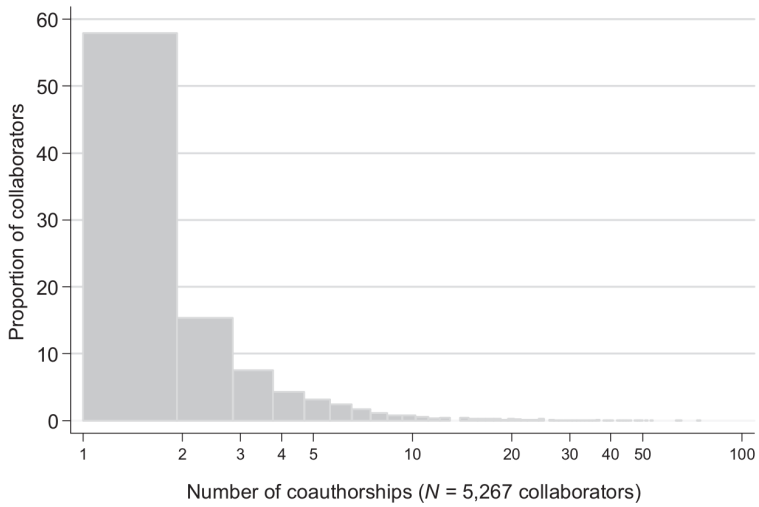

FIGURE I
Distribution of Coauthorship Intensity

Figure I: Distribution of Coauthorship Intensity.

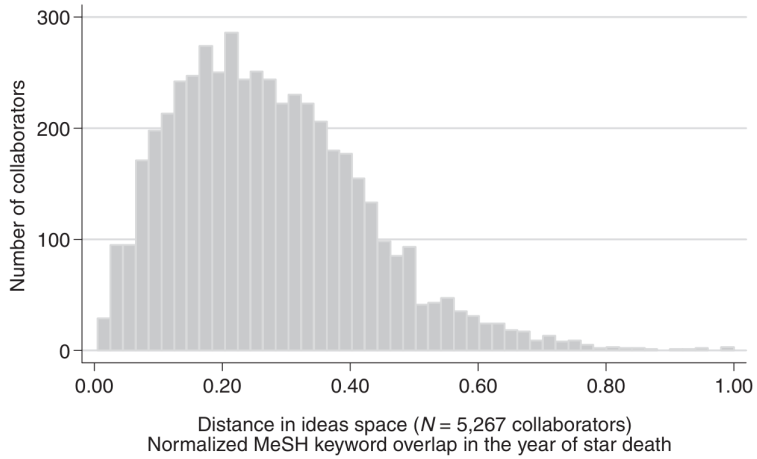

FIGURE II
Proximity in Ideas Space

Measure of distance in ideas space is defined as the number of unique MeSH terms that overlap between the colleague's and superstar's publications (excluding coauthored output), normalized by the total number of MeSH terms used in the colleague's total publications. This measure is calculated for articles published in the five years preceding superstar death. Calculation excludes coauthored publications.

Figure II: Proximity in Ideas Space.

6.4 Table III and Figure III: Main Effect and Dynamics

Read:

- Table III reports the core effect of superstar death on collaborator publication rates.
- Without matched controls, the post-death coefficient is -0.092, implying an 8.79% decline in all JIF-weighted publications.
- With matched controls, the coefficient is -0.086, implying an 8.24% decline.
- When publications written with the superstar are excluded, the decline remains statistically significant at about 5%.
- Figure III shows event-time dynamics. The treatment effect becomes more negative after death and appears persistent rather than temporary.

Economic message: The main result is not just a mechanical disappearance of joint star papers. Collaborators' broader publication output falls, and the effect does not quickly reverse.

TABLE III

IMPACT OF SUPERSTAR DEATH ON COLLABORATORS' PUBLICATION RATES

	Panel A All JIF-weighted publications		Panel B JIF-weighted publications written with others	
	Without ctrls (1a)	With ctrls (1b)	Without ctrls (2a)	With ctrls (2b)
After death	−0.092**	−0.086**	−0.057**	−0.054*
	(0.022)	(0.025)	(0.022)	(0.024)
Log pseudo-likelihood	−974,285	−1,832,594	−950,864	−1,783,958
No. of observations	153,508	294,943	153,508	294,943
No. of collaborators	5,267	10,128	5,267	10,128

Notes. Estimates stem from conditional quasi–maximum likelihood Poisson specifications. Dependent variable is the total number of JIF-weighted articles authored by a collaborator of a superstar life scientist in the year of observation. All models incorporate a full suite of year effects as well as seventeen age category indicator variables (career age less than −3 is the omitted category). Exponentiating the coefficients and differencing from one yield numbers interpretable as elasticities. For example. the estimates in column (1a) imply that collaborators suffer on average a statistically significant $(1 - \exp[-0.092]) = 8.79\%$ decrease in the rate of publication after their superstar coauthor passes away. Robust (QML) standard errors in parentheses, clustered at the level of the superstar.

$^{*}p < .05$.

$^{**}p < .01$.

Table III: Impact of Superstar Death on Collaborators' Publication Rates.

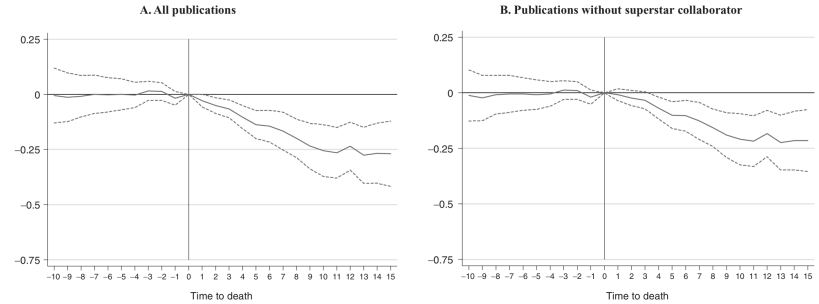

FIGURE III

Dynamics of the Treatment Effect

The solid lines in the above plots correspond to coefficient estimates of conditional fixed effects quasi–maximum likelihood Poisso
pecifications in which the weighted publication output of a collaborator is regressed onto year effects, seventeen indicator variable
orresponding to different age brackets, and interactions of the treatment effect with 27 indicator variables corresponding to eleven year
efore the year of death and prior, ten years before the year of death, nine years before the year of death,. . . , fourteen years after th
ear of death, and fifteen years after the year of death and above (the indicator variable for treatment status interacted with the yea
f death is omitted). The 95% confidence interval (corresponding to robust standard errors, clustered around superstars) around thes
stimates is plotted with dashed lines. The graph for all publications (A) uses column (1b) of Table III as a baseline (i.e., treated an
ontrol collaborators, the dependent variable, includes all of the collaborator's publications); the graph for publications without superst
ollaborator (B) uses column (2b) of Table III as a baseline (i.e., treated and control collaborators, the dependent variable, is limited t
e collaborator's publications in which the superstar does not appear on the authorship list).

Figure III: Dynamics of the Treatment Effect.

6.5 Table IV and Table V: Skill Substitution and Resource Access

Read:

- Table IV asks whether the decline is simply imperfect replacement of a missing team member.
- Regular, close, and recent collaborators do not experience statistically significant additional declines relative to casual or less recent collaborators.
- This weakens a pure skill-substitution interpretation.
- Table V tests resource-access and gatekeeping stories.
- The effects are not stronger for collaborators of stars tied to NIH review panels, stars with more NIH-panel coauthor ties, more central stars, or former trainees.

Economic message: The output decline is not well explained by losing a narrow production input or by losing access to funding and influence. Something broader than a replaceable team skill or gatekeeper role appears to disappear.

TABLE IV
COLLABORATOR PUBLICATION RATES AND IMPERFECT SKILL SUBSTITUTION

	Coauthorship intensity		Coauthorship recency		Coauthorship intensity & recency	
	All pubs. (1a)	Pubs. written with others (1b)	All pubs. (2a)	Pubs. written with others (2b)	All pubs. (3a)	Pubs. written with others (3b)
After death	−0.076**	−0.057*	−0.087**	−0.074**	−0.080**	−0.075**
	(0.026)	(0.025)	(0.024)	(0.024)	(0.024)	(0.024)
After death × regular collaborator	−0.044	−0.020			−0.039	−0.018
	(0.041)	(0.042)			(0.042)	(0.043)
After death × close collaborator	−0.026	0.117			−0.014	0.119
	(0.068)	(0.073)			(0.069)	(0.074)
After death × at least one coauthorship in the three years preceding star's death			−0.022	0.032	−0.021	0.028
			(0.038)	(0.039)	(0.039)	(0.039)
Log pseudo-likelihood	−1,831,987	−1,781,742	−1,822,664	−1,775,680	−1,821,791	−1,774,167
No. of observations	294,943	294,943	294,943	294,943	294,943	294,943
No. of collaborators	10,128	10,128	10,128	10,128	10,128	10,128

Notes. Estimates stem from conditional quasi–maximum likelihood Poisson specifications. Dependent variable is the total number of JIF-weighted articles authored by a collaborator of a superstar life scientist in the year of observation. Regular and close collaborator are indicator variables for the number of publications coauthored by the superstar and colleague at the time of death (regular collaborations correspond to between three and nine coauthored publications; close collaborations correspond to ten or more coauthored publications; casual collaborations—the omitted category—corresponds to one or two coauthored publications). All models incorporate year effects and seventeen age category indicator variables (career age less than −3 is the omitted category), as well as seventeen interaction terms between the age effects and each covariate of interest (i.e., column (3b) includes a total of $3 \times 17 = 51$ age-specific interaction terms). Robust (QML) standard errors in parentheses, clustered at the level of the superstar.
$^{*}p < .05$.
$^{**}p < .01$.

Table IV: Collaborator Publication Rates and Imperfect Skill Substitution.

TABLE V
COLLABORATOR PUBLICATION RATES AND ACCESS TO RESOURCES

	Star's ties to NIH funding process (1)	Quartile of betweenness centrality (2)	Former trainee (3)	All covariates combined (4)
After death	−0.105**	−0.067*	−0.086**	−0.089*
	(0.037)	(0.028)	(0.025)	(0.035)
After death × star sat on NIH review panel	0.042			0.024
	(0.064)			(0.070)
After death × star's no. of coauthor ties to NIH review panelists	0.011			0.014
	(0.013)			(0.015)
After death × star in fourth quartile of betweenness centrality		−0.031		−0.040
		(0.046)		(0.051)
After death × coauthor is former trainee			0.056	0.048
			(0.069)	(0.069)
Log pseudo-likelihood	−1,831,339	−1,831,779	−1,830,582	−1,828,754
No. of observations	294,943	294,943	294,943	294,943
No. of collaborators	10,128	10,128	10,128	10,128

Notes. Estimates stem from conditional quasi–maximum likelihood Poisson specifications. Dependent variable is the total number of JIF-weighted articles authored by a collaborator of a superstar life scientist in the year of observation. Betweenness centrality is measured using the network of 10,349 superstar life scientists, former trainee indicates that the colleague was a graduate student or postdoctoral fellow in the laboratory of the superstar (7.69% of the collaborators). All models incorporate year effects and seventeen age category indicator variables (career age less than −3 is the omitted category), as well as seventeen interaction terms between the age effects and each covariate of interest. Robust (QML) standard errors in parentheses, clustered at the level of the superstar.
$^{*}p < .05$.
$^{**}p < .01$.

Table V: Collaborator Publication Rates and Access to Resources.

6.6 Table VI and Table VII: Idea-Space Spillovers and Superstar Status

Read:

- Table VI studies physical and intellectual proximity.
- Colocation with the superstar does not predict a larger decline.
- High MeSH keyword overlap does predict a larger decline: collaborators in the top quartile of keyword overlap suffer an additional negative effect.
- Accidental collaborators appear much less affected, consistent with the idea that a meaningful social tie is needed to enter the star's invisible college.
- Table VII shows that collaborators of highly cited stars suffer larger declines, while collaborators of especially well-funded stars do not.

Economic message: The key spillover channel is intellectual rather than merely geographic or financial. Ideas are most valuable to collaborators working near the superstar's research frontier.

TABLE VI

Collaborator Publication Rates and Proximity in Geographic and Intellectual Space

	(1)	(2)	(3)	(4)	(5)
After death	−0.092**	−0.067**	−0.094**	−0.081**	−0.074**
	(0.027)	(0.023)	(0.022)	(0.024)	(0.026)
After death × colocated	0.042			0.037	0.042
	(0.043)			(0.043)	(0.044)
After death × kwd. overlap in top quartile		−0.115*		−0.114†	−0.127*
		(0.059)		(0.059)	(0.057)
After death × "accidental" collaborator			0.104†	0.111†	0.077
			(0.060)	(0.058)	(0.055)
After death × regular collaborator					−0.030
					(0.044)
After death × close collaborator					0.002
					(0.072)
After death × recent collaborator					−0.022
					(0.038)
% of collaborators affected	13.33	25.35	7.53		
Log pseudo-likelihood	−1,831,900	−1,830,305	−1,831,787	−1,828,805	−1,817,667
No. of observations	294,943	294,943	294,943	294,943	294,943
No. of collaborators	10,128	10,128	10,128	10,128	10,128

Notes. Estimates stem from conditional quasi–maximum likelihood Poisson specifications. Dependent variable is the total number of JIF-weighted articles authored by a collaborator of a superstar life scientist in the year of observation. Colocated indicates that the colleague and superstar were employed at the same institution at the time of superstar death. Keyword overlap is the normalized number of MeSH keywords that appear on both the colleague's and superstar's nonjoint publications. Accidental collaborators are those who only appear on coauthored publications with the superstar when both are in the middle of the authorship list. Regular and close collaborator are indicator variables for the number of publications coauthored by the superstar and colleague at the time of death (regular collaborations correspond to between three and nine coauthored publications; close collaborations correspond to ten or more coauthored publications; casual collaborations—the omitted category—correspond to one or two coauthored publications). All models incorporate year effects and seventeen age category indicator variables (career age less than −3 is the omitted category), as well as seventeen interaction terms between the age effects and the covariate of interest. Robust (QML) standard errors in parentheses, clustered at the level of the superstar.

†$p < .10$.

*$p < .05$.

Table VI: Collaborator Publication Rates and Proximity in Geographic and Intellectual Space.

TABLE VII

Impact of Superstar Status on Collaborators' Publication Rates

	Superstar status citations (1)	Superstar status NIH funding (2)	Superstar status citations & NIH funding (3)
After death	−0.034	−0.070*	−0.026
	(0.036)	(0.035)	(0.039)
After death × star in top quartile of cites	−0.082†		−0.080†
	(0.047)		(0.048)
After death × star in top quartile of NIH $		−0.026	−0.016
		(0.050)	(0.051)
Log pseudo-likelihood	−1,715,929	−1,716,213	−1,715,916
No. of observations	275,776	275,776	275,776
No. of collaborators	9,470	9,470	9,470

Notes. Estimates stem from conditional quasi–maximum likelihood Poisson specifications. Dependent variable is the total number of JIF-weighted articles authored by a collaborator of a superstar life scientist in the year of observation. Top quartiles of citations and career NIH funding are defined using the population of 10,009 superstar scientists with appointments compatible with extramural NIH funding. We exclude from the estimation sample the collaborators of eleven "intramural" NIH scientists who are *not* eligible to receive extramural funding. All models incorporate year effects and seventeen age category indicator variables (career age less than −3 is the omitted category), as well as seventeen (columns (1) and (2)) or 34 (column (3)) interaction terms between the age effects and the "top quartile" indicator variable. Robust (QML) standard errors in parentheses, clustered at the level of the superstar.

†$p < .10$.

*$p < -.05$.

Table VII: Impact of Superstar Status on Collaborators' Publication Rates.

6.7 Appendix Figures A.1 and A.2: Matching Diagnostics

Read:

- Figure A.1 plots publication trends for treated and control collaborators around the death event.
- Before the event, treated and control trends are similar; after the event, treated collaborators decline relative to controls.
- Figure A.2 plots differential age trends for treated versus control collaborators before superstar death.
- The estimates are centered near zero across most ages, supporting the use of the matched control group to absorb life-cycle patterns.

Economic message: These appendix figures support the identifying comparison. They show that the treated group does not simply have a different pre-existing trend relative to controls.

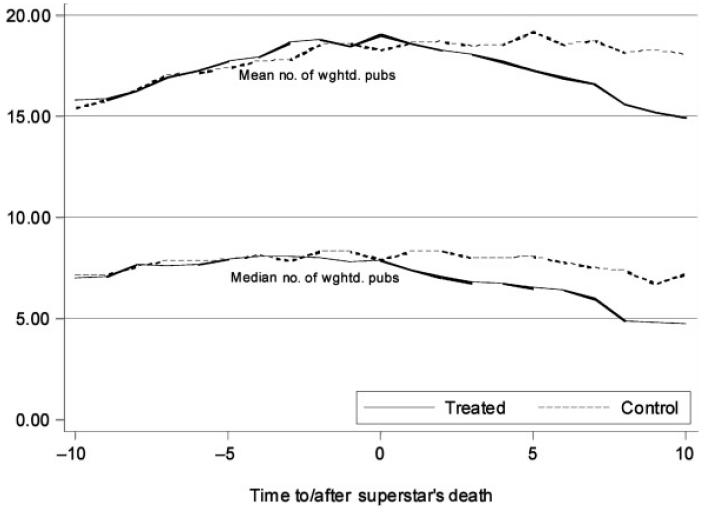

FIGURE A.1
Publication Trends for Treated and Control Collaborators

Figure A.1: Publication Trends for Treated and Control Collaborators.

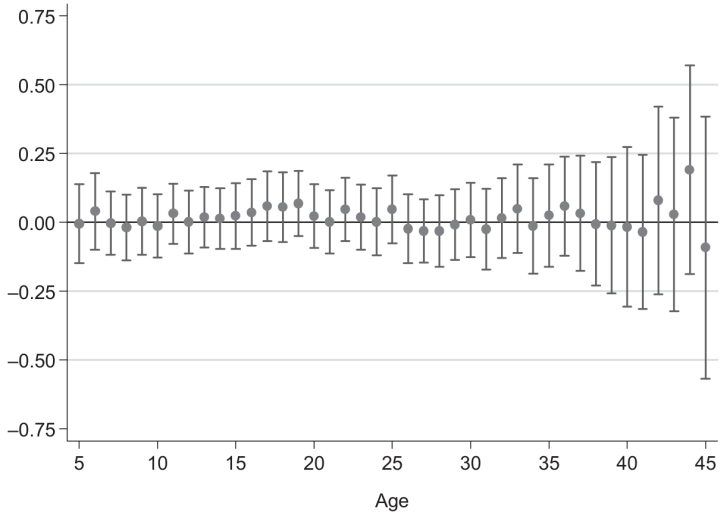

FIGURE A.2
Differential Age Trends for Treated vs. Control Collaborators

Each dot corresponds to the coefficient estimate for the interaction between an age indicator variable and treatment status in a Poisson regression of weighted publications onto a full suite of year effects, a full suite of age effects, and age by treatment status interaction terms. The population includes control and treated collaborators, but the estimation sample is limied to the years *before* the death of

Figure A.2: Differential Age Trends for Treated vs. Control Collaborators.

7 Short summary

- The paper estimates the spillovers generated by superstar scientists using sudden and unexpected deaths as quasi-experimental shocks.
- The treatment sample contains 112 superstar life scientists who die while still research-active.
- The authors identify 5,267 treated collaborators and match 5,064 of them to control collaborators.
- The main outcome is JIF-weighted publication output.
- After a superstar dies, collaborators experience a lasting 5% to 8% decline in publication rates.
- The decline remains when publications coauthored with the superstar are excluded.
- Close and recent collaborators do not suffer statistically significant additional losses, so imperfect skill substitution within ongoing teams does not fully explain the result.
- NIH-review ties, network centrality, and former trainee status do not strongly predict larger declines, weakening gatekeeping explanations.
- Collaborators intellectually close to the star suffer larger declines, and collaborators of highly cited stars suffer larger declines.
- The main conclusion is that superstar scientists generate knowledge spillovers that are bounded mainly by idea space.

8 Conclusion

8.1 Core conclusion

Azoulay, Graff Zivin, and Wang show that superstar scientists generate substantial spillovers for collaborators. When an active superstar dies suddenly and unexpectedly, surviving coauthors experience a persistent decline in quality-adjusted publication output. The magnitude is economically meaningful: roughly 5% to 8% fewer JIF-weighted publications after the loss of the superstar collaborator.

8.2 Economic intuition

The intuition is that scientific collaboration transmits more than labor inputs. A superstar can supply ideas, direction, tacit knowledge, field-level vision, and intellectual momentum. These inputs are hard to replace because they are embedded in the research frontier and in the star's invisible college. When the superstar disappears, collaborators do not merely lose a coauthor; they lose access to a source of scientific imagination and problem selection.

8.3 Incremental contribution revisited

The paper advances the literature in three ways. First, it introduces a clean empirical shock - sudden superstar death - to study collaboration spillovers. Second, it builds a detailed individual-level panel

linking elite scientists, collaborators, publications, citations, grants, and network characteristics. Third, it uses heterogeneity across dyads and stars to distinguish mechanisms. The strongest evidence points to idea-based spillovers rather than resource gatekeeping or simple team-skill replacement.

8.4 Implications for science policy

The results imply that the social value of an eminent scientist can exceed the scientist's own publication record. If collaborators become more productive by working near a star, then grant agencies, universities, and research teams should think about the spillover structure of science. Funding an extraordinary scientist may also support an invisible college around that scientist. Conversely, the death or departure of a superstar can impose costs on a field that standard individual productivity metrics miss.

8.5 Limitations

The design estimates the effect of losing a superstar collaborator, not the full lifetime value of the superstar's presence. The setting is academic life sciences, so the results may not generalize mechanically to fields with different collaboration norms, different capital intensity, or less transparent publication records. Mechanism tests are based on heterogeneity patterns rather than randomized variation in mechanisms. Some indirect spillovers to noncoauthors are also outside the main sample.

8.6 Takeaway

The enduring takeaway is that scientific productivity is partly relational. Superstars matter not only because they publish influential work themselves, but because they help define and energize the research agenda of nearby collaborators. In this paper, the boundaries of that influence are not mainly institutional or geographic; they are intellectual.